

HAN XIAO

Contact Information

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Employment

Assistant Professor of Finance, Chinese University of Hong Kong, Shenzhen, 2023 to present

Education

PhD., Finance, Pennsylvania State University, 2018 to 2023
M.A., Economics, Pennsylvania State University, 2018
M.A., Economics, Renmin University of China, with distinction, 2018
B.A., Economics, Nanjing University, with distinction, 2014

Research Interests

Mutual funds, ETFs, hedge funds, ESG, and derivatives

Publications

Retail ETF Investing, with David Gempesaw and Joseph Henry, forthcoming, European Financial Management, 2024

Abstract: Using marketable order flow data, we analyze key characteristics of aggregate retail exchange-traded fund (ETF) investing from 2010 to 2021, including allocations, holding period and investment performance. Retail traders allocate 12% more dollar volume to leveraged and inverse ETFs versus nonretail traders. Retail ETF trades distinctly increase with prior ETF returns, in contrast to contrarian stock trading. Estimated ETF holding periods are longer for retail investors versus nonretail. Finally, retail and nonretail ETF trades perform similarly over hypothetical holding periods up to one quarter. Overall, we provide policy-relevant insights into retail investing behaviours, which have been the subject of recent concern.

Predicting Equity Premium with Implied Volatility Spreads, with Charles Cao and Timothy Simin, Journal of Financial Markets 51 (Lead Article), 2020

Abstract: We show that the call-put implied volatility spread (*IVS*) outperforms many well-known predictors of the equity premium at return horizons up to 6-months. The predictive ability of the *IVS* is unrelated to the dividend yield and is useful in explaining the cross-section of returns. Decomposing the *IVS*, we find the longer run predictive ability of the *IVS* operates primarily through a cash flow channel. We also find the *IVS* is significantly related to indicators of aggregate market direction and expected market conditions. Our results are consistent with the *IVS* reflecting market sentiment as well as information about informed trading.

Presentation: MFA, CICF, AFA Ph.D. Poster Session, Penn State University, CMU/Pitts/Penn State/OSU conference*, Renmin University of China, PBFEM*
(* Presented by coauthors)

Reprint: Chapter 10 in *Handbook of Investment Analysis, Portfolio Management, and Financial Derivatives* (2022), C. F. Lee, Editor, World Scientific

The Fragility of Financial System Under the Growing Shadow Banking System in China, with Lin Lin and Yong Cao, China Economic Quarterly, 2016 (3), pp. 1113-1136. (in Chinese)

Abstract: The paper establishes a DSGE model by including a shadow banking system to study the impact of growing shadow banking system in China on the Chinese economy. On the analysis of the enterprise financing behavior, financial risk accumulation and transformation mechanism, the relationship between the real economy fluctuation and these under the Chinese Shadow Banking System, the paper studied the financial system fragility. The empirical result shows that during economic booming fueled by the abnormal social credit expansion through largely unregulated shadow banking activities, enterprises tend to be over-financing which forms the basic condition of financial crisis. The sensitive relationship between enterprises' increasing financial leverage and the profitability of shadow bank system is the predetermined condition for financial crisis. Such internal fragility of the financial system influenced by the growing shadow bank system is prone to face substantial risk of credit crunch.

Working Papers

From Conference Room to Living Room: The Hidden Costs of Remote Work on Mutual Fund Performance, with Charles Cao and Timothy Simin, Mar. 2024 (Submitted)

Abstract: We use staggered U.S. state-level stay-at-home COVID-19 pandemic orders to study the effect of remote work on actively managed mutual fund performance. After implementation of working from home mandates, daily fund returns decrease by 90 basis points per day, corresponding to a daily \$6 million economic loss per fund relative to investing in the market portfolio. Remote work decreases measures of managerial skill, especially for funds not within a family and funds which are non-team managed. Mandated working from home during the COVID-19 pandemic period impaired mutual fund performance by increasing frictions which hinder idea sharing and information flows.

CCF Best Paper Award

Academy of Finance Best Paper Award in Investment

Reprinted at European PMC

Presentation: International Risk Management Conference, CCF, Penn State University, SWFA, EasternFA, SGF Ph.D. Poster, FMCG, GFC, CMES, ESAM, AsianFA, CIRF, SFA, After the Pandemic conference, Academy of Finance, University of North Texas*

(* Presented by coauthors)

Navigating Inflation Risk in Corporate Bond Markets, with Luis Ceballos, Mar. 2024 (Submitted)

Abstract: The global inflation surge has refocused attention on the impact of inflation risks. We investigate whether mutual fund managers time the inflation risks in the corporate bond market. Our findings reveal a significant and robust timing ability among managers in different investment subcategories, translating into a sizable fund performance of around 4% per annum. Timing is associated with managers adjusting portfolio holdings to bet on future risks rather than past realizations. Cross-sectional evidence suggests that over 40% individual funds exhibit strong inflation risk timing ability, controlling alternative timing abilities, factor structures, and monetary policy shocks. The bootstrapping exercise further validates managerial skills rather than pure luck. Our results provide policy implications for monetary policy transmission in corporate bond markets.

Presentation: SBFC, BAR, SWFA*, California Corporate Finance Conference*

Liquidity transformation in ETF Creation and Redemption, with Charles Cao and Hong Zhang, Jan. 2024

Abstract: This paper investigates the effectiveness of the redemption mechanism in the corporate bond exchange-traded funds (ETFs) market under liquidity mismatch. I begin with incentives underlying the choice of bonds in redemption. In the primary market between ETF sponsors and authorized participants (APs), sponsors fill baskets with bonds exposed to price pressure to eliminate arbitrage opportunities. APs select assets that negatively correlate with illiquidity in their overall bond portfolios to diversify liquidity risks. In the secondary market, where ETFs are publicly traded, traders carry the costs of redemption through a high price multiplier of shares and an illiquid and price-inefficient market.

Small Research Grant, 2022

Presentation: Penn State University, NAMES, AFR International Symposium, ESAM, CIRF, SETA, BFGA, AFBC, AFA PhD Poster, Inter-Finance PhD Seminar Series, PhD EVS, EasternFA, FMCG PhD, HK PolyU Fixed Income and Institutions Research Symposium

Media Mentions: FinReg Blog at the Global Financial Markets Center Duke University School of Law, Smeal Updates

Feedback, Flow-induced Fire Sales, and Option Returns, Mar. 2022

Abstract: We identify a feedback loop between fire sales and equity option returns. The demand effect of fire sales induced by mutual fund extreme outflows decreases delta-hedged put option returns by 4 - 10% per year and increases the expensiveness by 2.5%. The results hold in the instrumental variable and difference-in-differences designs addressing identification concerns. The demand effect inherits from equity illiquidity rather than volatility, distress, sustainability risks, or short-sale constraints. The put option returns also have anticipation effects on predicting fire sales, suggesting information leakage in derivatives markets exacerbates extreme outflows.

FMCG Best Paper Runner-up Award in Asset Pricing

Presentation: Penn State University, FMA New Idea Session, AFBC, NZFM, WFBS, CFRIC, Nova SBE, SWFA, Academy of Finance, CMU/PITT/PSU, EasternFA, FMCG Ph.D. Form, AsianFA, FMA, SFA

Variance Risk Premiums and Resolution of Uncertainty, with Charles Cao and Timothy Simin, Feb. 2020

Abstract: This paper empirically investigates the relationship between variance risk premiums (*VRP*) and the resolution of uncertainty using implied volatility. We show that the early resolution of uncertainty contributes to the *VRP* and the difference between the S&P index and individual firm *VRP*. The term structure of individual firm's *VRP* is downward sloping and is related to the early resolution of uncertainty. The early resolution of uncertainty is associated with higher *VRP* around macroeconomic and firm-specific events. These results support the early resolution of uncertainty assumptions in macro-finance models.

Small Research Grant, 2020

Best Paper Award in Derivative Semifinalist, FMA, 2020

Presentation: FMA, Penn State University

Working in Progress

Mutual Funds and ESG, with Sean Cao, Yong Chen, and Alan L. Zhang

Discussion

SBFC (2023, Sydney): Security Lending Market, Secondary Market Arbitrageurs, and ETF Mispricing, Wu.

BAR (2023, Boston): Foreign Signal Radar, Jiao.

CICF (2023, Shanghai): The Persistent Response from Option Liquidity to GameStop Short Squeeze, Yang and Zhong.

Ad hoc referee

Journal of Behavioral and Experimental Finance, Journal of Finance and Data Science, China Economic Quarterly

Teaching Experience

Fall 2023

Alternative Investment, CUHK-Shenzhen

Fall 2023

Financial Markets in China and the World, CUHK-Shenzhen

Summer, Fall 2021

Derivatives, Penn State

Summer 2020 Introduction to Corporate Finance, Penn State

Service

SBFC, 2023	Session chair
SFA, 2022	Committee member
FMA, 2021	Organize hedge fund session and paper review, Session chair
NZFM, 2021	Session chair

Languages/Skills/Others

Languages	Mandarin (native), English (Fluent)
Computers	SAS, Stata, MATLAB, R, Python
Other Qualification	CFA II Candidate